

# Xinai Zhang

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## EDUCATION

**Stanford University**  
*M.S. in Mechanical Engineering*

Stanford, CA  
*Expected May 2028*

**Brown University**  
*Sc.B. (Honors) in Biomedical Engineering — GPA: 4.0/4.0*  
Tau Beta Pi Engineering Honor Society Member — Top 1/8 of Junior Class and Top 1/5 of Senior Class

Providence, RI  
*Sep 2022 - May 2026*

## WORK EXPERIENCE

**Coulombe Lab, Brown University School of Engineering**

Sep 2025 – May 2026

*Undergraduate Researcher*

*Providence, RI*

- Developed a pulsatile stretch tissue-conditioning bioreactor system that conducted up to 40% stretch on engineered tissue and enhanced the functional performance of engineered bioelectric grafts for post-myocardial infarction cardiac repair
- Developed full mechanical design and CAD models, and wrote Arduino C++ firmware with TMC2209 microstepping control and UART real-time communication for high-precision linear actuation
- Specified and integrated biocompatible materials into a modular bioreactor architecture, meeting sterilization and biocompatibility requirements for tissue-contacting components
- Calibrated displacement and velocity control via motion capture ( $R^2 = 0.9924$ ), enabled 4 programmable stretch waveforms (uniform, asymmetric, trapezoidal, incremental) and validated system stability (amplitude CV<1% in 48 hrs)

**Cornerstone Robotics**

Jun 2025 – Aug 2025

*Clinical Development Engineer Intern*

*Hong Kong*

- Conducted clinical development research on curved bipolar dissectors; analyzed 230+ adverse event reports in the MAUDE database and identified 8+ design-specific risk factors; produced a comprehensive risk analysis report delivered to the R&D team to guide product iteration and inform downstream clinical strategy
- Supported 20+ porcine, canine, and cadaveric robotic-assisted surgical studies as circulating nurse and intraoperative assistant, managing pre-operative preparation and intraoperative coordination; documented pre-surgical clinical presentations and product performance issues, validated preclinical trial outcomes for the latest robot model
- Led development of 3 regulatory SOPs; built a streamlined project management and financial reporting system, and oversaw clinical documentation organization and progress tracking across concurrent workstreams

## PROJECT EXPERIENCE

**Ergonomic Endoscope Redesign**

Sep 2025 – May 2026

*Capstone Project Member*

*Providence, RI*

- Interviewed practicing endoscopists, observed live endoscope procedures, and tested a simulator firsthand to pinpoint finger strain as the core usability problem; proposed a slider-based actuation design adopted as the team's design direction
- Designed and CAD-modeled the slider housing and ergonomic palm-fitting outer shell, iterating through multiple 3D-printed prototypes to refine fit and mechanical function
- Designed and prototyped an adjustable handle mechanism with modular joints offering 180° of rotational freedom, allowing users to customize the handle to their hand size and grip
- Conducted a patent landscape analysis to map existing IP and confirm the design was meaningfully differentiated from prior solutions, informing its path toward market viability

**Muscle-Controlled Mario Kart with Electromyography Sensors**

Sep 2024 – Jan 2025

*Project Lead*

*Providence, RI*

- Developed a multi-stage EMG signal amplifier that converts muscle activity into control input for a Mario Kart emulator
- Developed signal-processing logic in Arduino (C++); calibrated discrimination threshold in LabVIEW for precise control

**Redesigning NASA Space Suit through Biomechanical Interventions**

Jan 2024 – May 2024

*Project Team Member*

*Providence, RI*

- Modeled astronaut gait in OpenSim to design a passive elastic assist for space suits, addressing fatigue and injury issues
- Developed biomechanical testing protocols to quantify assistive torque and improve suit mobility
- Built and tested a neoprene-elastomer wearable prototype, validated up to 8 Nm of torque assistance
- Presented findings at NASA Rhode Island Space Grant Consortium

## SKILLS

**Engineering Software:** Fusion 360, SolidWorks, MATLAB, LabVIEW, ImageJ, Arduino, OpenSim, Python, GraphPad

**Laboratory:** Cell Culture, Cell Assays, Animal Models, Western Blotting, Genotyping, Mechanical Testing, Soldering

**Digital Media & Design:** DaVinci Resolve, Adobe Premiere Pro, Final Cut Pro, Adobe Photoshop, Adobe InDesign